Problem Set 2

Econometrics

Code: https://github.com/ae4292/Econometrics_PSets

Problem 1

(a) Regression of lwage on the 30 education × age dummies (898 observations). Estimates $\hat{\alpha}_{e,a}$ with standard errors in parentheses:

	$a = 26$	$a = 27$	$a = 28$	$a = 29$	$a = 30$
$e = 12$	9.451 (0.091)	9.300 (0.085)	9.538 (0.083)	9.386 (0.085)	9.628 (0.088)
$e = 13$	9.612 (0.192)	9.756 (0.192)	9.378 (0.175)	9.820 (0.224)	9.446 (0.203)
$e = 14$	9.581 (0.224)	9.794 (0.171)	9.642 (0.216)	9.712 (0.175)	9.645 (0.171)
$e = 15$	9.535 (0.242)	9.386 (0.232)	9.573 (0.265)	9.540 (0.252)	9.919 (0.242)
$e = 16$	10.070 (0.167)	9.944 (0.187)	9.992 (0.148)	10.179 (0.131)	10.071 (0.164)
$e = 17$	9.958 (0.483)	10.046 (0.224)	10.166 (0.296)	9.618 (0.418)	10.016 (0.374)

α is the individual effect of having a given age and education on log wage.

(b)

$$\begin{aligned}
&\alpha_{e,a-1} - 2\alpha_{e,a} + \alpha_{e,a+1} = 0, && e = 12,\dots,17,\ a = 27,28,29,\\
&(\alpha_{e-1,27} - \alpha_{e-1,26}) - 2(\alpha_{e,27} - \alpha_{e,26}) + (\alpha_{e+1,27} - \alpha_{e+1,26}) = 0, && e = 13,\dots,16,\\
&\alpha_{e-1,26} - 2\alpha_{e,26} + \alpha_{e+1,26} = 0, && e = 13,\dots,16,
\end{aligned}$$

which gives R of size 26×30 and $q = 0$ (code in the appendix).

$$F = 0.920, \qquad p\text{-value} = 0.581$$

I would not reject the null hypothesis as the p-value (0.58) is not below 0.05 significance level.

Problem 2

(a)

$$\mathbb{E}(x_i x_i') = \frac{1}{n}\sum_{i=1}^{n} x_i x_i' = \frac{1}{n}\cdot\left(0 + 2^2 + 0 + 4^2 + \cdots + n^2\right) = \frac{1}{n}\cdot\frac{2n(n+1)(2n+1)}{3} < \infty$$

(sum of even numbers squared). $\mathbb{E}(\epsilon_i \mid x_i)$ is always 0.

$$\mathrm{Var}(y_i \mid x_i) = \mathrm{Var}(\epsilon_i \mid x_i)$$

$$\mathbb{E}(\epsilon_i x_i) = \mathbb{E}\big(\mathbb{E}(\epsilon_i x_i \mid x_i)\big) = \mathbb{E}_x\big(x_i\,\mathbb{E}(\epsilon_i \mid x_i)\big) = \mathbb{E}(x_i \cdot 0) = 0$$

$$\mathrm{Var}(\epsilon_i x_i) = \mathbb{E}(\epsilon_i^2 x_i^2) - \mathbb{E}(\epsilon_i x_i)^2 = \mathbb{E}\big(\mathbb{E}(\epsilon_i^2 x_i^2 \mid x_i)\big) = \mathbb{E}\big(x_i^2\,\mathbb{E}(\epsilon_i^2 \mid x_i)\big) = \mathbb{E}(x_i^2) < \infty$$

$\sigma^2\mathbb{E}(x_i^2)^{-1}$ shrinks as $n \to \infty$. OLS $\hat{\beta}$ for β is consistent, $\mathbb{E}[\hat{\beta}] = \beta$.

(b)

$$\mathbb{E}(x_i x_i') = \mathbb{E}(\lambda^{2i}) = \frac{1}{n}\sum \lambda^{2i} = \frac{\lambda^2}{n(1-\lambda^2)} < \infty$$

if $n \to \infty$, $\mathbb{E}x_i x_i' = 0$ so it is singular.

$$\mathbb{E}(x_i x_i')^{-1} = \frac{n(1-\lambda^2)}{\lambda^2}$$

this however is never converging as $n \to \infty$, thus $\hat{\beta}$ does not converge in probability to β as the variance will be infinite as $n \to \infty$.

(c) For 2a we can have perfect collinearity since for $n = 1$, $\text{rank}(X) \neq 1$. However for 2b Gauss-Markov holds. Both have strict exogeneity, homoskedasticity and no serial correlation.

(d)

$$\begin{aligned}
\sqrt{n}(\hat{\beta}_{OLS} - \beta) &\to N\big(0, \sigma^2\mathbb{E}(x_i x_i')^{-1}\big) \\
&\to N\left(0, \sigma^2\frac{n(1-\lambda^2)}{\lambda^2}\right) \quad \text{from 2b} \\
\hat{\beta}_{OLS} &\to N\left(\beta, \sigma^2 \cdot \frac{1-\lambda^2}{\lambda^2}\right), \qquad \sigma^2 = 1
\end{aligned}$$

> $$\hat{\beta}_{OLS} \to N\left(\beta, \frac{1-\lambda^2}{\lambda^2}\right)$$

Problem 3

(a) $\theta = \dfrac{s_1^2}{\sigma^2} = 1, \quad g(\theta) = \log(\theta), \quad g'(\theta) = \dfrac{1}{\theta} = 1.$

$$(\hat{\theta} - \theta) \sim N\big(0, (g(\theta))^2 \cdot V\big) = N(0, 1 \cdot V) = N(0, 2)$$

$$\frac{s_1^2}{\sigma^2} = \frac{\chi^2_{n-1}}{n-1}, \qquad \text{Var}\left(\frac{s_1^2}{\sigma^2}\right) = \text{Var}\left(\frac{\chi^2_{n-1}}{n-1}\right) = \frac{2n}{n-1} \xrightarrow[n\to\infty]{} 2$$

$$\begin{pmatrix} \sqrt{n_1}\log\dfrac{s_1^2}{\sigma^2} \\ \sqrt{n_2}\log\dfrac{s_2^2}{\sigma^2} \end{pmatrix} = N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\right) \qquad \text{independent samples no cov.}$$

$$\log(s_1^2) - \log(s_2^2) = \log(s_1^2) - \log(\sigma^2) - \log(s_2^2) + \log(\sigma^2) = \log\left(\frac{s_1^2}{\sigma^2}\right) - \log\left(\frac{s_2^2}{\sigma^2}\right)$$

$$T = \sqrt{\frac{n_2}{n_1+n_2}} \cdot \sqrt{n_1}\log\left(\frac{s_1^2}{\sigma^2}\right) - \sqrt{\frac{n_1}{n_1+n_2}}\sqrt{n_2}\log\left(\frac{s_2^2}{\sigma^2}\right)$$

$$\mathbb{E}(T) = \sqrt{\frac{n_2}{n_1+n_2}} \cdot \mathbb{E}\left(\sqrt{n_1}\log\frac{s_1^2}{\sigma^2}\right) - \sqrt{\frac{n_1}{n_1+n_2}}\,\mathbb{E}\left(\sqrt{n_2}\log\frac{s_2^2}{\sigma^2}\right) = 0$$

$$\mathrm{Var}(T) = \frac{n_2}{n_1+n_2}\,\mathrm{Var}\left(\sqrt{n_1}\log\frac{s_1^2}{\sigma^2}\right) + \frac{n_1}{n_1+n_2}\,\mathrm{Var}\left(\sqrt{n_2}\log\frac{s_2^2}{\sigma^2}\right) = 2\cdot\frac{n_2+n_1}{n_1+n_2} = 2$$

(b) $P(T \geq z_{0.95}\sqrt{2}) = 0.05$

$$F(n_1-1, n_2-1) = \frac{s_1^2/(n_1-1)}{s_2^2/(n_2-1)} \text{ as } n_1, n_2 \to \infty \;\to\; \frac{s_1^2}{s_2^2}\cdot c \sim \frac{s_1^2}{s_2^2}$$

$P(F > q_{0.95})$ at large n

$$\Rightarrow P\left(\frac{s_1^2}{s_2^2} > q_{0.95}\right) = P\left(\sqrt{\frac{n_1 n_2}{n_1+n_2}}\log\frac{s_1^2}{s_2^2} > \sqrt{\frac{n_1 n_2}{n_1+n_2}}\log(q_n)\right)$$

$$P\left(T > \sqrt{\frac{n_1 n_2}{n_1+n_2}}\log(q_n)\right) = 0.05$$

This is equivalent to $P(T > z_{0.95}\sqrt{2}) = 0.05$ as they have same probability as long as n_1 and $n_2 \to \infty$.

(c)

$$\mathbb{E}(T) = \sqrt{\frac{n_2}{n_1+n_2}}\,\mathbb{E}\left(\sqrt{n_1}\log\frac{s_1^2}{\sigma^2}\right) - \sqrt{\frac{n_1}{n_1+n_2}}\,\mathbb{E}\left(\sqrt{n_2}\log\frac{s_2^2}{\sigma^2}\right)$$

$$y_{ji} = \mu + \sigma z_{ji}$$

$$s_1^2 = \frac{1}{n_1-1}\sum\left(\mu+\sigma z_{ji} - \frac{1}{n_1}\sum\mu+\sigma z_{ji}\right)^2 = (\sigma z_{ji} - \sigma\bar{z}_j)^2 = \sigma^2\sum(z_{ji}-\bar{z}_j)^2\Big/(n_1-1)$$

$$\frac{s_1^2}{\sigma^2} = \frac{\sum(z_{ji}-\bar{z}_j)^2}{n_1-1} = \frac{\sum z_{ji}^2 - n_1\bar{z}_j^2}{n_1-1} = \frac{n_1}{n_1-1}\left(\frac{1}{n_1}\sum z_i^2 - \bar{z}^2\right)$$

$\frac{s_1^2}{\sigma^2}$ as $n_1 \to \infty$: $\bar{z} = \frac{1}{n_1}\sum z_i$, $\frac{1}{n}\sum z_i^2 \to 1$ as $\int y^2\,dF_0(y) = 1$

$$\frac{n_1}{n_1-1}\left(1-\frac{1}{n_1}\right) = \frac{n_1-1}{n_1-1} = 1, \qquad \frac{s^2}{\sigma^2} \xrightarrow{p} 1$$

$$\sqrt{n_1}\left(\frac{s_1^2}{\sigma^2}-1\right) = \sqrt{n_1}\left(\frac{n_1}{n_1-1}\Big(\frac{1}{n_1}\sum z_i^2 - \bar{z}^2\Big) - 1\right) \quad \text{as } n_1 \to \infty$$

$$\sqrt{n}\,\frac{n}{n+1}\left(\frac{1}{n}\sum z_i^2 - \bar{z}^2 - 1\right) = \sqrt{n}\Big(\frac{1}{n}\sum z_i^2 - \bar{z}^2\Big) + \frac{1}{n+1}\Big(\sum z_i^2 - \bar{z}^2\Big) - 1$$

$$= \sqrt{n}\,\frac{1}{n}\sum z_i^2 - \sqrt{n} - \sqrt{n}\,\bar{z}^2 + \frac{\sqrt{n}}{n+1}\Big(\sum z_i^2 - \bar{z}^2\Big)$$

$\sqrt{n}\,\bar{z}^2 \to \sqrt{n}\cdot 0 \to 0$ as n increases

$$\frac{\sqrt{n}}{n+1}\Big(\sum z_i^2 - \bar{z}^2\Big) = \frac{\sqrt{n}}{n+1}\cdot 1 - 0 = \frac{\sqrt{n}}{n+1} \quad \text{which} \to 0 \text{ as } n \text{ increases}$$

thus $\mathbb{E}\left(\sqrt{n}\,\frac{s_1^2}{\sigma^2}-1\right) = \sqrt{n}\Big(\frac{1}{n}\sum z_i^2 - 1\Big) \to \sqrt{n}(1-1) = 0$

$$\mathrm{Var}\left(\sqrt{n}\,\frac{s_1^2}{\sigma^2}-1\right) = \mathrm{Var}\left(\frac{1}{\sqrt{n}}\sum z_i^2 - 1\right) = \frac{1}{n}\,\mathrm{Var}\left(\sum z_i^2\right) = \frac{1}{n}\cdot n(\kappa-1)$$

$$\text{Var}\left(\sqrt{n}\,\frac{s_1^2}{\sigma^2}-1\right)=\kappa-1$$

delta method

$$\sqrt{n}\log\left(\frac{s_1^2}{\sigma^2}-1\right)=\sqrt{n}\log\left(\frac{s_1^2}{\sigma^2}\right)\to N\left(0,\left(\frac{\partial g(\theta)}{\partial\theta}\right)^2(\kappa-1)\right)$$

$$\left(\frac{\partial g(\theta)}{\partial\theta}\right)^2=\left(\frac{1}{\theta}\right)^2=\left(\frac{1}{1}\right)^2=1$$

$$\sqrt{n_1}\log\left(\frac{s_1^2}{\sigma^2}\right)\Rightarrow N(0,\kappa-1)$$

$$\mathbb{E}(T)=\sqrt{\frac{n_2}{n_1+n_2}}\cdot 0+\sqrt{\frac{n_2}{n_1+n_2}}\cdot 0=0$$

$$\text{Var}(T)=\left(\frac{n_1}{n_1+n_2}+\frac{n_2}{n_1+n_2}\right)(\kappa-1)=\kappa-1$$

(d)

$$P(T>1.645\sqrt{2})=P\left(Z>\frac{1.645\cdot\sqrt{2}}{2}\right)=0.122$$

we have a 12.2% chance of rejecting the null if $\kappa=5$ is the true distribution.

(e)

> p-value robust to non-normality $=0.175$, p-value under normality $=0.047$

The p-value under normal assumptions shows that we would reject the null with 0.05 significance level, while the non-normality robust test shows we would not reject the null.

Problem 4

(a)

$$\sqrt{n}(\hat{\beta}-\beta)\xrightarrow{d}N\left(0,\sigma^2(\mathbb{E}(xx'))^{-1}\right)$$

$\sigma^2=\text{Var}(\epsilon_i\mid j_i)=1$ stated in problem.

$$\mathbb{E}(xx')^{-1}\Rightarrow\mathbb{E}(j_ij_i')^{-1}=\left(\frac{1}{p}\sum j_ij_i'\right)^{-1}\Rightarrow p\cdot 1$$

j is indicator which has $\mathbb{E}(I^2)=\mathbb{E}(I)=1/p$.

$$\sqrt{n}(\hat{\beta}-\beta)\xrightarrow{d}N(0,p)$$

(b)

$$\min\ \|y-Xb\|^2+\lambda\sum|b_j|$$

$$\|y-Xb\|^2=\sum_j^p\sum_i^{n_j}(y_i-b_j)^2=\sum_j^p\sum_i^{n_j}(y_i-\hat{\beta}_j)^2+\sum_j^p n_j(\hat{\beta}_j-b_j)^2=\sum_j^p n_j(\hat{\beta}_j-b_j)^2+\sum\epsilon'\epsilon$$

($\sum \epsilon'\epsilon$ can't minimize with b_j)

$$\min \sum_j^p n_j(\hat{\beta}_j - b_j)^2 + \lambda \sum |b_j|$$

FOC b_j: $\quad -2n_j(\hat{\beta}_j - b_j) \pm \lambda = 0 \qquad 2n_j\hat{\beta}_j = 2n_j b_j \pm \lambda$

$$b_j = \hat{\beta}_j \pm \frac{\lambda}{2n_j}$$

$$b_j^{LASSO} = \begin{cases} \hat{\beta}_j + \dfrac{\lambda}{2n_j} & \hat{\beta}_j < -\dfrac{\lambda}{2n_j} \\ \hat{\beta}_j - \dfrac{\lambda}{2n_j} & \hat{\beta}_j > \dfrac{\lambda}{2n_j} \\ 0 & \text{otherwise} \end{cases}$$

(c) $\lambda = \sqrt{n}$

$$\mathbb{E}\big(\sqrt{n}(\hat{\beta}_{Lasso} - \beta)\big) = \sqrt{n}\Big(\hat{\beta}_j - \frac{\sqrt{n}}{2n_j} - \beta\Big) = -\sqrt{n} \cdot \frac{\sqrt{n}}{2n_j} = -\frac{n}{2n_j} = -\frac{p}{2}$$

$\frac{n}{n_j} = \frac{1}{p}$, $\quad p$: chance we select j: $\frac{n_j}{n} = \frac{A_j}{\text{all aliens}} = p$

$$\text{Var}\left(\sqrt{n}(\hat{\beta}_{Lasso} - \beta)\right) = \text{Var}\left(\sqrt{n}(\hat{\beta}_j - \beta)\right) - \text{Var}\left(\sqrt{n} \cdot \frac{\sqrt{n}}{2n_j}\right) \to \text{Var}\left(\sqrt{n}(\hat{\beta}_j - \beta)\right) = p$$

the second term $\to 0$.

$$\sqrt{n}(\hat{\beta}_{Lasso} - \beta) \xrightarrow{d} N\Big(-\frac{p}{2}, p\Big)$$

$\lambda = n^{1/4}$

Var is unchanged but expectation will change

$$\mathbb{E}\big(\sqrt{n}(\hat{\beta}_{Lasso} - \beta)\big) = -\sqrt{n} \cdot \frac{n^{1/4}}{2n_j} = \frac{n^{3/4}}{2n_j} = \frac{p}{2} \cdot \frac{1}{n^{1/4}} \to 0$$

$$\sqrt{n}(\hat{\beta}_{Lasso} - \beta) \xrightarrow{d} N(0, p)$$

$\lambda = n^{3/4}$

$$\mathbb{E}\big(\sqrt{n}(\hat{\beta}_{Lasso} - \beta)\big) = -\sqrt{n} \cdot \frac{n^{3/4}}{2n_j} = \frac{n^{5/4}}{2n_j} = n^{1/4} \cdot \frac{p}{2} \to \infty$$

$$\sqrt{n}(\hat{\beta}_{Lasso} - \beta) \xrightarrow{d} N(\infty, p)$$

(d) $\hat{\beta}_{Lasso} = 0$ when $\hat{\beta} \leq \left|\frac{\lambda}{2n_j}\right| = \left|\frac{\sqrt{n}}{2n_j}\right| = \left|\frac{p}{2}\right|$

$$P\Big(-\frac{\sqrt{n}}{2n_j} \leq \hat{\beta} \leq \frac{\sqrt{n}}{2n_j}\Big) = P\Big(-\frac{p}{2} \leq \sqrt{n}\hat{\beta} \leq \frac{p}{2}\Big)$$

$\sqrt{n}(\hat{\beta} - \beta) \to \sqrt{n}(\hat{\beta})$ as $\beta = 0$ if useless alien

for having k useless aliens we need $\left(P\left(-\frac{p}{2} \leq \sqrt{n}\hat{\beta} \leq \frac{p}{2}\right)\right)^k$

$$P\left(-\frac{p}{2} \leq \sqrt{n}\hat{\beta} \leq \frac{p}{2}\right) = \left(2\Phi\left(\frac{p}{2} \cdot \frac{1}{\sqrt{p}}\right) - 1\right) = 2\Phi\left(\frac{\sqrt{p}}{2}\right) - 1$$

Φ is normal cdf

> asymptotic probability of k useless aliens is $\left(2\Phi\left(\frac{\sqrt{p}}{2}\right) - 1\right)^k$

Problem 5

No. Strict stationarity means the joint distribution of a sequence doesn't depend on time t. The sequence can have auto-correlation and heteroskedasticity in this case. The distribution of (y_t, x_t) is constant, but if we condition the distribution on (y_{t-1}, x_{t-1}) then the distribution on $y_t \mid x_t$ can vary. Stationarity only enforces that the variance-covariance matrix doesn't depend on t, but it can depend on $t-1$ information if conditioned on it.

Appendix: Code

Setup

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import statsmodels.api as sm
```

Problem 1(a)

```
wages = pd.read_csv("ps1small.csv")

wages['logwage'] = np.log(wages['wage'])
wages['eduxage'] = 'd'+ wages['education'].astype(str) + wages['age'].astype(str)
dummies = pd.get_dummies(wages['eduxage'], dtype = int)
wages = pd.concat([wages, dummies], axis=1)
wages.head()
```

```
model = sm.OLS(endog= wages['logwage'], exog=wages[dummies.columns]).fit()
coefficients = model.params
standard_errors = model.bse
alpha = pd.DataFrame({'Coefficient': coefficients, 'Standard Error': standard_errors})
alpha
```

Problem 1(b)

```
R  = np.zeros((26,30))
q = np.zeros((26,1))

def col(e, a):
    # column index for education level e and age a
    return (e - 12) * 5 + (a - 26)

#alpha_{e,a-1} - 2*alpha_{e,a} + alpha_{e,a+1} = 0
r = 0
for e in range(12, 18):
    for a in (27, 28, 29):
        R[r, col(e, a-1)] = 1
        R[r, col(e, a)]   = -2
        R[r, col(e, a+1)] = 1
        r += 1

#(alpha_{e-1,27} - alpha_{e-1,26}) - 2*(alpha_{e,27} - alpha_{e,26}) + (alpha_{e+1,27} -
    alpha_{e+1,26}) = 0
for r in range(18, 22):
    e = 13 + (r - 18)
    R[r, col(e-1, 27)] = 1
    R[r, col(e-1, 26)] = -1
    R[r, col(e, 27)]   = -2
    R[r, col(e, 26)]   = 2
    R[r, col(e+1, 27)] = 1
    R[r, col(e+1, 26)] = -1

#alpha_{e-1,26} - 2*alpha_{e,26} + alpha_{e+1,26} = 0
for r in range(22, 26):
    e = 13 + (r - 22)
    R[r, col(e-1, 26)] = 1
    R[r, col(e, 26)]   = -2
    R[r, col(e+1, 26)] = 1
```

```
wald_test = model.wald_test((R,q), use_f = True)
F = wald_test.fvalue
p_value = wald_test.pvalue
print("F-statistic:", F)
print("p-value:", p_value)
```

Problem 3(d)

```
from scipy import stats
kappa = 5
# T ~ N(0, kappa-1) under the null, but the test uses the N(0,2) critical value 1.645*sqrt
    (2)
size = 1 - stats.norm.cdf(1.645*np.sqrt(2) / np.sqrt(kappa-1))
print(f'actual asymptotic size when kappa=5: {size:.4f}')
```

Problem 3(e)

```
from scipy import stats
```

```
wage = pd.read_excel('wage.xlsx')
```

```
wage['logwage'] = np.log(wage['wage'])
wage.head()
```

```
wage_males = wage[wage['male']==1]
wage_females = wage[wage['male']==0]

n1 = len(wage_females)
n2 = len(wage_males)

y1 = wage_females['logwage']
y2 = wage_males['logwage']
y1_bar = np.mean(y1)
y2_bar = np.mean(y2)

s1 = np.sum((y1-y1_bar)**2) / (n1-1)
s2 = np.sum((y2-y2_bar)**2) / (n2-1)

T = np.sqrt(n1*n2/ (n1+n2)) * (np.log(s1) - np.log(s2))

k_hat = (n1+n2)* (np.sum((y1-y1_bar)**4) + np.sum((y2-y2_bar)**4)) / (np.sum((y1-y1_bar)
    **2) + np.sum((y2-y2_bar)**2)) **2

z = T / np.sqrt(k_hat-1)
p_robust = 1- stats.norm.cdf(z)

F = s1/s2
p_normal = 1 - stats.f.cdf(F,n1-1,n2-1)

print(f'p-value robust to non-normality {p_robust}')
print(f'p value under normality {p_normal}')
```